

Arithmetic Operations
Complements
Data Representation

TABLE 1-1
Numbers with Different Bases

Decimal (base 10)	Binary (base 2)	Octal (base 8)	Hexadecimal (base 16)
00	0000	00	0
01	0001	01	1
02	0010	02	2
03	0011	03	3
04	0100	04	4
05	0101	05	5
06	0110	06	6
07	0111	07	7
08	1000	10	8
09	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Examples

Convert 1001011_2 to decimal.

Convert $(0.513)_{10}$ to octal.

Convert 52_{10} to binary

Convert $A0F9.0EB_{16}$ to decimal.

Find base

$$\frac{41}{3} = 13$$

Convert 1001011_2 to decimal.

$$1001011_2 = 75_{10}$$

Convert $(0.513)_{10}$ to octal.

$$(0.513)_{10} = (0.406517 \dots)_8$$

Convert 52_{10} to binary

$$52_{10} = 110100_2.$$

Convert $A0F9.0EB_{16}$ to decimal.

$$41209.0572_{10}$$

Find base

$$\frac{41}{3} = 13$$

base 8 system.

Binary Number System

Addition

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 10$$

EXAMPLE 2.13 Add the binary numbers 1101.101 and 111.011.

Solution

	8	4	2	1		2^{-1}	2^{-2}	2^{-3}	(Column numbers)
	1	1	0	1	.	1	0	1	
+		1	1	1	.	0	1	1	
	1	0	1	0	1	.	0	0	0

In the 2^{-3} 's column $1 + 1 = 0$, with a carry of 1 to the 2^{-2} column
In the 2^{-2} 's column $0 + 1 + 1 = 0$, with a carry of 1 to the 2^{-1} column
In the 2^{-1} 's column $1 + 0 + 1 = 0$, with a carry of 1 to the 1's column
In the 1's column $1 + 1 + 1 = 1$, with a carry of 1 to the 2's column
In the 2's column $0 + 1 + 1 = 0$, with a carry of 1 to the 4's column
In the 4's column $1 + 1 + 1 = 1$, with a carry of 1 to the 8's column
In the 8's column $1 + 1 = 0$, with a carry of 1 to the 16's column.

Subtraction

$$0 - 0 = 0$$

$$0 - 1 = 1 \text{ (borrow of 1)}$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

EXAMPLE 2.15 Subtract 111.111_2 from 1010.01_2 .

Solution

8	4	2	1		2^{-1}	2^{-2}	2^{-3}	(Column numbers)
1	0	1	0	.	0	1	0	
	1	1	1	.	1	1	1	
0	0	1	0	.	0	1	1	

Multiplication

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

EXAMPLE 2.17 Multiply 1011.101_2 by 101.01_2 .

Solution

$$\begin{array}{r} 1011.101 \\ \times 101.01 \\ \hline 1011101 \\ 0000000 \\ 1011101 \\ 0000000 \\ 1011101 \\ \hline 11110100001 \end{array}$$

Octal Number System

Addition

EXAMPLE 2.47 Add 327.54_8 to 665.37_8 .

Solution

8^3	8^2	8^1	8^0	8^{-1}	8^{-2}	
	3	2	7	.	5	4
+	6	6	5	.	3	7
1	2	1	5	.	1	3

In the 8^{-2} 's column, $4 + 7 = 11_{10} = 13_8$ (sum 3, carry 1)
In the 8^{-1} 's column, $5 + 3 + 1 = 9_{10} = 11_8$ (sum 1, carry 1)
In the 8^0 's column, $7 + 5 + 1 = 13_{10} = 15_8$ (sum 5, carry 1)
In the 8^1 's column, $2 + 6 + 1 = 9_{10} = 11_8$ (sum 1, carry 1)
In the 8^2 's column, $3 + 6 + 1 = 10_{10} = 12_8$ (sum 2, carry 1)

Subtraction

EXAMPLE 2.48 Subtract 16.47_8 from 20.14_8 .

Solution

$$\begin{array}{rcccccc} & 8^1 & 8^0 & & 8^{-1} & 8^{-2} \\ & 2 & 0 & . & 1 & 4 \\ - & 1 & 6 & . & 4 & 7 \\ \hline & 0 & 1 & . & 4 & 5 \end{array}$$

Hexadecimal Number System

Addition

EXAMPLE 2.66 Add $3BCA.5078_{16} + 9EBD.97F3_{16} + 5FB.E2C_{16}$.

Solution

	3	B	C	A	.	5	0	7	8
+	9	E	B	D	.	9	7	F	3
		5	F	B	.	E	2	C	
	E	0	8	3	.	C	B	2	B

Subtraction

EXAMPLE 2.67 Subtract $78D6.3B_{16}$ from $B08E.A1_{16}$.

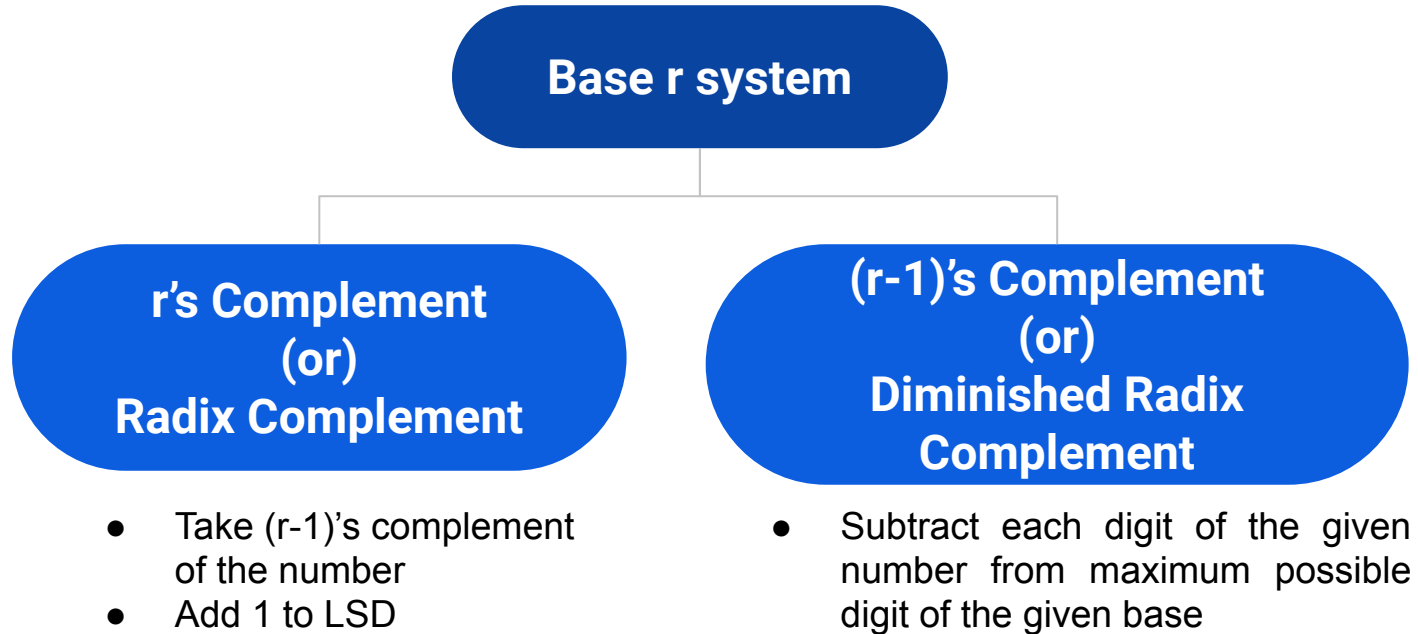
Solution

The hex subtraction is performed as explained below.

	16^3	16^2	16^1	16^0	.	16^{-1}	16^{-2}
	B	0	8	E	.	A	1
–	7	8	D	6	.	3	B
	3	7	B	8	.	6	6

Complements

- For simplifying the Subtraction Operation
- For logical manipulation
- Complement of the complement restores the number to its original value



Example of (r-1)'s Complement

9's Complement of 3465

$$\begin{array}{r} 9999 \\ - 3465 \\ \hline 6534 \end{array}$$

1's Complement of 1011

$$\begin{array}{r} 1111 \\ - 1011 \\ \hline 0100 \end{array}$$

F's Complement of 2675

$$\begin{array}{r} FFFF \\ - 2675 \\ \hline D98A \end{array}$$

Example of r's Complement

10's Complement of 3465

$$\begin{array}{r} 9999 \\ - 3465 \\ \hline 6534 \\ + \quad 1 \\ \hline 6535 \end{array}$$

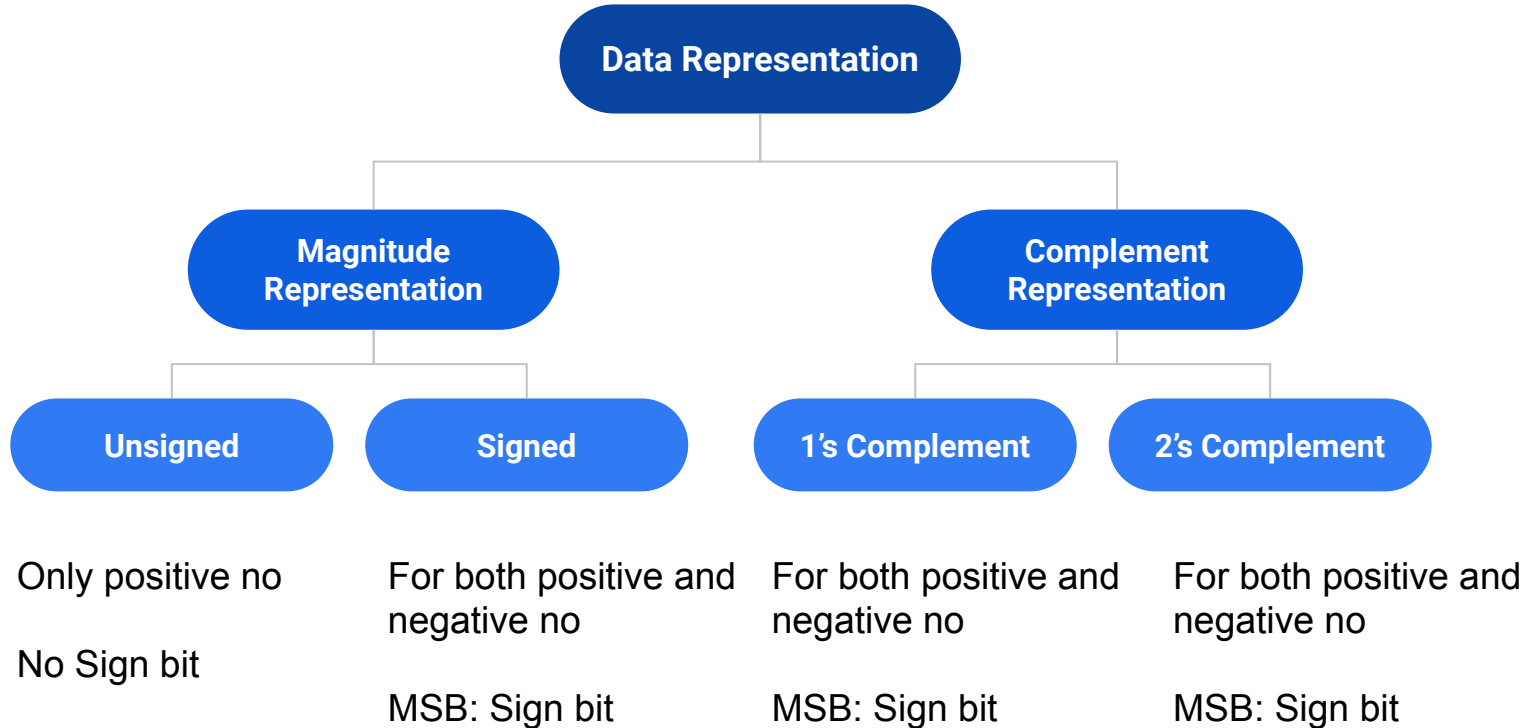
2's Complement of 1011

$$\begin{array}{r} 1111 \\ - 1011 \\ \hline 0100 \\ + \quad 1 \\ \hline 0101 \end{array}$$

16's Complement of 2675

$$\begin{array}{r} FFFF \\ - 2675 \\ \hline D98A \\ + \quad 1 \\ \hline D98B \end{array}$$

Data Representation



Unsigned Numbers

- Positive numbers only
- Range of Integers: 0 to $(2^n - 1)$
- Eg: $n = 4$: 0 to 15

0000	0	1000	8
0001	1	1001	9
0010	2	1010	10
0011	3	1011	11
0100	4	1100	12
0101	5	1101	13
0110	6	1110	14
0111	7	1111	15

Signed Numbers

- Both positive and negative numbers
- Leftmost bit: Sign bit
- Sign bit: 0 $\rightarrow$ Positive number +13 : 01101
- Sign bit: 1 $\rightarrow$ Negative number -13 : 11101
- Range of Integers: $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$
- Eg: n = 4: -7 to +7

0000	+0	1000	-0
0001	+1	1001	-1
0010	+2	1010	-2
0011	+3	1011	-3
0100	+4	1100	-4
0101	+5	1101	-5
0110	+6	1110	-6
0111	+7	1111	-7

Disadvantage: 2 zeros

1's Complement

- For both positive and negative numbers
- Positive number : Normal binary form +13 : 01101
- Negative number : 1's Complement form -13 : 10010
- Range of Integers : $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$
- Eg: n = 4 : -7 to +7

0000	+0	1000	-7
0001	+1	1001	-6
0010	+2	1010	-5
0011	+3	1011	-4
0100	+4	1100	-3
0101	+5	1101	-2
0110	+6	1110	-1
0111	+7	1111	-0

Disadvantages:

- 2 zeros
- Addition: if carry is present it is added to LSB to get correct result

$$X - Y = 1010100 - 1000011$$

$$\begin{array}{r}
 X = \quad \quad 1010100 \\
 1\text{'s complement of } Y = \quad + \underline{0111100} \\
 \text{Sum} = \quad \quad 10010000 \\
 \text{End-around carry} \quad \quad \quad \rightarrow + \underline{1} \\
 \text{Answer: } X - Y = \quad \quad 0010001
 \end{array}$$

$$Y - X = 1000011 - 1010100$$

$$\begin{array}{r}
 Y = \quad \quad 1000011 \\
 1\text{'s complement of } X = \quad + \underline{0101011} \\
 \text{Sum} = \quad \quad 1101110
 \end{array}$$

There is no end carry.

$$\text{Answer: } Y - X = -(1\text{'s complement of } 1101110) = -0010001$$

EXAMPLE 2.33 Subtract 14 from 25 using the 8-bit 1's complement arithmetic.

Solution

$$\begin{array}{r}
 25 \\
 - 14 \\
 \hline
 + 11
 \end{array}
 \Rightarrow
 \begin{array}{r}
 00011001 \\
 + 11110001 \\
 \hline
 \textcircled{1} 00001010 \\
 + 1 \\
 \hline
 00001011
 \end{array}
 \begin{array}{l}
 \text{(In 1's complement form)} \\
 \text{(Add the end around carry)} \\
 = + 11_{10}
 \end{array}$$

EXAMPLE 2.34 Add -25 to $+14$ using the 8-bit 1's complement method.

Solution

$$\begin{array}{r} +14 \\ -25 \\ \hline -11 \end{array} \Rightarrow \begin{array}{r} 00001110 \\ +11100110 \\ \hline 11110100 \end{array} \begin{array}{l} \text{(In 1's complement form)} \\ \text{(No carry)} \end{array}$$

There is no carry and the MSB is a 1. So, the result is negative and is in its 1's complement form. The 1's complement of 11110100 is 00001011. The result is, therefore, -11_{10} .

EXAMPLE 2.36 Add + 25 to + 14 using the 8-bit 1's complement arithmetic.

Solution

$$\begin{array}{r} + 25 \\ + 14 \\ \hline + 39 \end{array} \Rightarrow \begin{array}{r} 00011001 \\ + 00001110 \\ \hline 00100111 \end{array}$$

There is no carry. The MSB is a 0. So, the result is positive and is in pure binary. Therefore, the result is, $00100111 = + 39$.

2's Complement

- For both positive and negative numbers
- Positive number : Normal binary form +13 : 01101
- Negative number : 2's Complement form -13 : 10011
- Range of Integers : $-(2^{n-1})$ to $(2^{n-1} - 1)$
- Eg: n = 4 : -8 to +7

0000	+0	1000	-8
0001	+1	1001	-7
0010	+2	1010	-6
0011	+3	1011	-5
0100	+4	1100	-4
0101	+5	1101	-3
0110	+6	1110	-2
0111	+7	1111	-1

Given the two binary numbers $X = 1010100$ and $Y = 1000011$, perform the subtraction (a) $X - Y$ and (b) $Y - X$ using 2's complements.

$$\begin{array}{rcl} \text{(a)} & X = & 1010100 \\ & 2\text{'s complement of } Y = & + \underline{0111101} \\ & \text{Sum} = & 10010001 \\ & \text{Discard end carry } 2^7 = & - \underline{10000000} \\ & \text{Answer: } X - Y = & 0010001 \end{array}$$

$$\begin{array}{rcl} \text{(b)} & Y = & 1000011 \\ & 2\text{'s complement of } X = & + \underline{0101100} \\ & \text{Sum} = & 1101111 \end{array}$$

There is no end carry.

$$\text{Answer: } Y - X = -(2\text{'s complement of } 1101111) = -0010001$$

EXAMPLE 2.25 Subtract 14 from 46 using the 8-bit 2's complement arithmetic.

Solution

$$\begin{array}{rcl}
 + 14 & = & 00001110 \\
 - 14 & = & 11110010 \text{ (In 2's complement form)} \\
 \hline
 + 46 & & 00101110 \\
 - 14 & \Rightarrow & + 11110010 \text{ (2's complement form of } -14) \\
 \hline
 + 32 & & \textcircled{0}0010000 \text{ (Ignore the carry)}
 \end{array}$$

There is a carry, ignore it. The MSB is 0; so, the result is positive and is in normal binary form. Therefore, the result is $+0010000 = +32$

EXAMPLE 2.26 Add -75 to $+26$ using the 8-bit 2's complement arithmetic.

Solution

$$\begin{array}{rcl}
 + 75 & = & 01001011 \\
 - 75 & = & 10110101 \text{ (In 2's complement form)} \\
 \hline
 + 26 & & 00011010 \\
 - 75 & \Rightarrow & + 10110101 \text{ (2's complement form of } -75) \\
 \hline
 - 49 & & 11001111 \text{ (No carry)}
 \end{array}$$

There is no carry, the MSB is a 1. So, the result is negative and is in 2's complement form. The magnitude is 2's complement of 11001111, that is, $00110001 = 49$. Therefore, the result is -49 .

Binary Arithmetic

$$\begin{array}{r} + 6 \quad 00000110 \\ +13 \quad 00001101 \\ \hline +19 \quad 00010011 \end{array}$$

$$\begin{array}{r} + 6 \quad 00000110 \\ -13 \quad 11110011 \\ \hline - 7 \quad 11111001 \end{array}$$

$$\begin{array}{r} - 6 \quad 11111010 \\ +13 \quad 00001101 \\ \hline + 7 \quad 00000111 \end{array}$$

$$\begin{array}{r} - 6 \quad 11111010 \\ -13 \quad 11110011 \\ \hline -19 \quad 11101101 \end{array}$$

Overflow

+5 : 0 1 0 1

+4 : 0 1 0 0

1 0 0 1 -7

-5 : 1 0 1 1

-4 : 1 1 0 0

0 1 1 1 +7

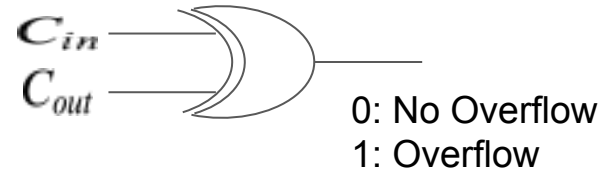
Detection of Overflow:

1. Using Sign bit

X	Y	Z	Overflow Flag
0	0	1	1
1	1	0	1
1	0	0	0
0	1	1	0

$$\text{Condition} = X'Y'Z + XYZ'$$

2. Using Carry



Extend number of bits

- In 1's and 2's complement representation, MSB is copied into extra position

		Signed	1's Comp	2's Comp
4-bit	+5	0101	0101	0101
	-5	1101	1010	1011
8-bit	+5	00000101	00000101	00000101
	-5	10000101	11111010	11111011